
How to Change the Pickups on an Electric Guitar



Tools You Will Need to Complete This Tutorial:

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| ■ Towel/Pillow | ■ Soldering Iron | ■ Screwdriver Set |
| ■ Wire Strippers | ■ Pliers | ■ Electrical Tape |
| ■ Guitar Strings | ■ Guitar Cable | ■ Amplifier |
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NEW AND EXPERIENCED PLAYERS KNOW THAT GUITAR PRICES RANGE WIDELY. For instance, a Fender-knockoff, most notably Squire, sells under \$250 CAD, amp and gig-bag included. Gibson, on the other hand, rarely falls below the \$1000 CAD threshold. In fact, some high-end guitars cost more than a semester's tuition.

Many things determine price—wood, finishing, design and electronics. Therefore, as a budget-conscious student, you need to learn how to make upgrades that will transform an otherwise cheap guitar into a decent-sounding axe. Besides simple changes like strings or straps, swapping pickups is one of the easiest and most impactful modifications available.

Generally, you can change the pickups on your guitar in under one hour. It may, however, take longer the first time. Note that the process also requires some electrical skills, primarily when stripping and soldering wires. For this reason, read the instructions carefully and be mindful of the cautions along the way.

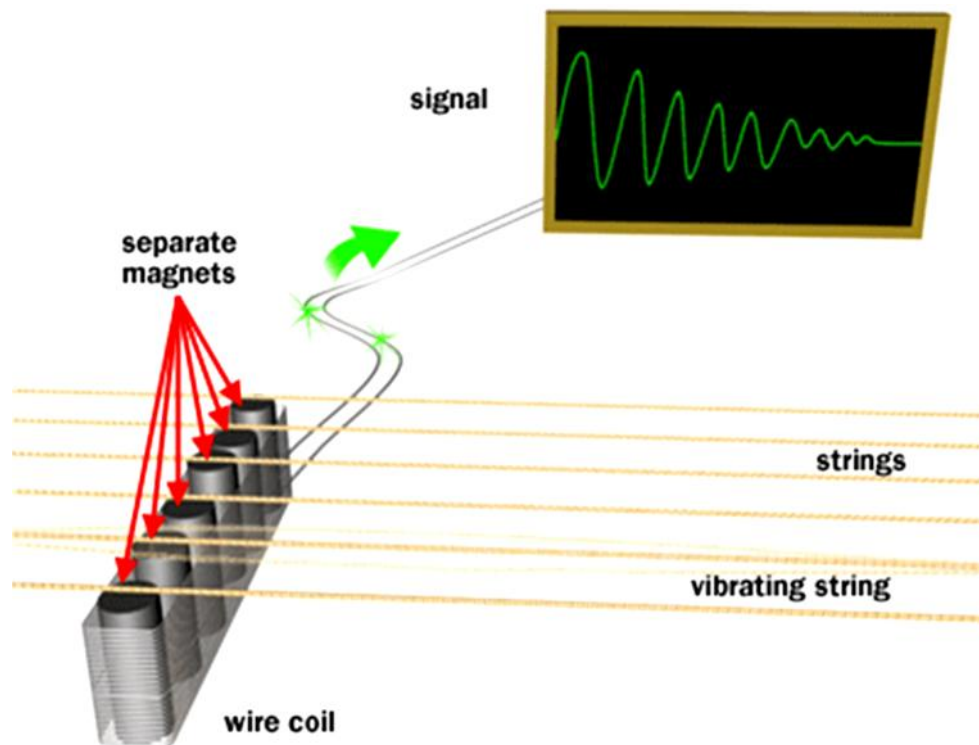


Figure 1: Diagram of a single-coil electric guitar pickup (Credit How Stuff Works)

What are Pickups?

Electric guitars use magnetic transducers, known commonly as *pickups*, to convert the vibrating strings into signals playable through an amp (Figure 1).

Generally, music stores sell two types of pickups: single- and double-coil. Double-coil pickups, or *humbuckers*, use two wires to insulate the magnets and reduce feedback. They play louder and thicker than single-coil pickups, though some guitarists prefer a brighter, crisper sound. In this regard, many argue that single-coil pickups define notes more clearly.

Expect to spend about one hundred dollars on new pickups. Always sample a few before purchasing because they greatly impact sound, playability and value.

Prepping the Pickups

Before opening your guitar, prep the pickups. If you own a vintage pair, you will notice two wires: a

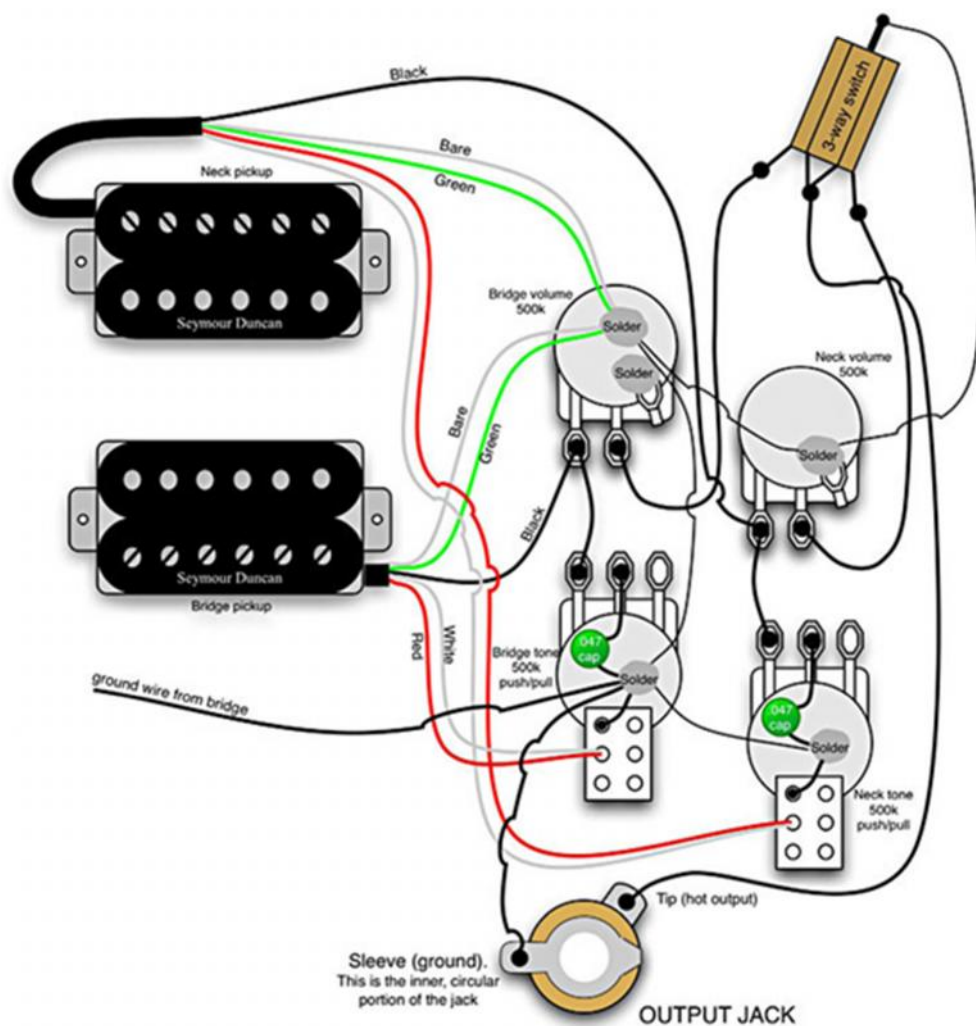


Figure 2: Schematics for a Seymour Duncan pickup on a four-knob guitar.

ground wire (bare) and a *hot* connection. Hot refers to the wire's ability to produce sound.

Modern pickups come with one insulated wire encasing five slimmer wires: two ground wires (one sheathed, one bare), one hot connection and two others. This tutorial focuses on modern installations; however, it also covers a simple setup and thus does little with the two other wires.

Moving forward, consult the wiring diagram—typically provided by the manufacture—to find each wire's corresponding colour (Figure 2). Get them ready by completing the following:

1. Strip an inch of plastic sheathing off each wire;
2. Braid and solder the grounding wires together, and then do the same to the two other wires (see page 6);
3. Tape the exposed, soldered wires afterwards.

Installation Walkthrough

Lay the new pickups down next to your guitar. Keep them at a convenient distance to ensure the installation goes smoothly.

To protect your instrument, prop the neck up with a pillow and rest the body on a towel. If you own a proper guitar mat, use it instead. To avoid complications, unplug the guitar at this time too. Leaving the amp on damages the speaker; heavy feedback is unavoidable.

Remove the strings before flipping the guitar onto its back. Do this by slackening them from the headstock and snipping them near the bridge. Should you wish to preserve your strings, unwrap gently and place them aside. You can also take this opportunity to clean the fretboard.

Removing Old Pickups

Turn the guitar over to access the back electrical plate. Unscrew the cover, but careful not to lose the screws in the guitar or on the floor. Reference your diagram to navigate the innards (i.e. the different wires and bulbs). Bulbs, sometimes called *pots*, refer to the active ends of tone and volume controls. Once comfortable with the setup, perform the steps below:

1. Flip the guitar over once again to remove the screws holding your old pickups in place;
2. Tug on the loose pickups while looking at the back bulbs to see where each wire connects;
3. Use pliers to grasp each wire and safely de-solder from its respective bulb (see page 6);
4. Disconnect the wires and pull the pickups from the front.

Connecting New Pickups

After pulling the pickups out, quickly grab your new pair. If you did not prep already, return to pages 2 and 3. Should you work swiftly, the solders may still be in liquid form, enabling you to simply dip the new wires and avoid additional soldering. With that said, practice caution as to not damage the instrument nor the pickups during the stages below:

1. Slide the wires through the same holes you pulled the old pickups from;
2. Attach the wires to the bulbs, recalling the last configuration and consulting the schematics (use the soldering iron again if necessary);
3. Pull gently on the wires after a few seconds to confirm they are attached.

Before screwing the new pickups into place, test the installation by plugging the guitar into an amp, turning up the volume and tapping the new pickups with a screwdriver. A popping noise means everything's working. Continue to tap as you turn the various tone and volume controls on your guitar, ensuring each works properly.

To complete the installation, screw on the new pickups, replace the electrical cover and re-string the guitar. Tune and crank up the amp, enjoying the new pickups you've installed! Be sure to test the knobs again from time-to-time over the following week to confirm the wires haven't fallen out of place.

If you've enjoyed this tutorial, you may wish to explore other installation methods such as *coil-tapping* or *coil-splitting*. These are complex techniques for advanced players, making use of the two other wires mentioned briefly in this lesson.

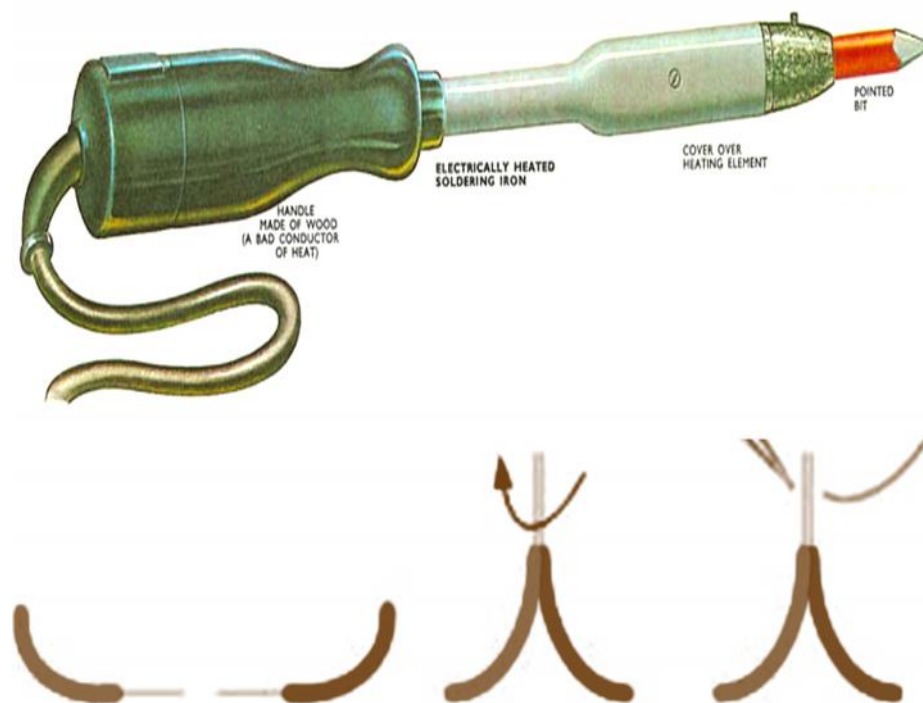


Figure 3: Pencil-based Soldering Iron with Wire/Solder Diagram (Credit: David Darling and Beavis Audio).

How to Solder Wires

Caution: Do not inhale the fumes and avoid touching the iron's bit; it's hot! Always use pliers to protect your fingers.

Begin by stripping the wire (see pages 2 and 3) and preheating the soldering iron. Once ready, place the wire between the bit and the solder—a thin, fusible alloy used to join metals together (Figure 3). Verify that the solder is touching the appropriate bulb, then gently push the bit into the wire. Heat the wire enough to melt the solder: it will smoke and droop between 2 to 20 seconds.

Do not touch the bit of the iron to the solder itself. Doing so creates a *cold solder joint*, which makes for a poor connection. You do not want to create a puddle of solder either.

Once finished, secure the iron appropriately in its stand or holder, turn off and move it away from the station where you are working.